

Systems Modeling

Systems Modeling and Its Processes

- A **model** depicts a conceptual framework to describe a system and can be viewed as an abstraction of an actual or a physical facsimile of a system or a situation.

Commonly used abstracted model

- Logical model** – It is a model that represents all the plausible requirements of a system. It can be combined with or independent from a physical model.
- Mathematical model** – It is a mathematical description of properties and interactions in a system. Its development depends on the system boundaries, components, interactions, type of analysis that will be performed, and the assumption that will be considered during the development process.

The mathematical model usually describes a system using *variables*. Variables can be real numbers, integers, Boolean values, strings, or practically anything that can represent a system's specific property, such as signals, timing data, counters, and event occurrences.

According to how much prior information is available for the system, mathematical modeling problems are often classified as white-box or black-box models. A *white-box* model encompasses systems where all necessary information is available, and a *black-box* model is associated with systems without prior information available. However, in reality, most systems fall between the two categories. Thus, this concept is only applicable as an intuitive guide approach (Chaturvedi, 2010).

- Models are practically used to mimic the behavior of a system under different operating conditions. The approximation is inevitable in order to develop a more robust and simpler model.
- Models developed based on a system theory paradigm bear a resemblance to concepts of object-oriented programming since both objects and system models share a concept of internal state (Zeigler, Muzy & Kofman, 2019).
- Modeling** pertains to the process of abstraction of an actual system, either existing or for development. The purpose of modeling is to depict internal processes and present as useful to science and engineering studies.
- Various input requirements and modeling methodology may be used in systems modeling, depending upon the purpose and an analyst's viewpoint.

- The mathematical model of a system component embodies its behavior as an independent entity, showing how it is interconnected with the other components to form a system successfully. It implies that various components can be removed either literally or conceptually and studied in isolation to establish a model of their characteristics. Analysts can go far in breaking down systems, searching for the building blocks that would be enough to develop a model. (Chaturvedi, 2010).

Characteristics of A Model

There are numerous ways to describe the word model, as well as the process of modeling. However, in terms of system specifications, modeling has a strong advantage of having a sound mathematical foundation and definitive semantics that all stakeholders can understand (Zeigler, Muzy & Kofman, 2019).

Model Type	Descriptive Capability	Ambiguity	Manipulation Capability	Implementation Capability	Primary Function
English Text	Good	Very ambiguous	None	Limited	Descriptive explanations and directions
Drawings and Block Diagrams	Good	Not ambiguous	None	Good	Design, assembly, and construction
Logical Flow Charts and Decision Tables	Fair	Not ambiguous	None	Good	Computer programming
Curves, Tables, and Monographs	Fair	Not ambiguous	Good	None	Expresses relationships between few variables
Mathematical Models	Poor	Not ambiguous	Excellent	Good	Problem solving and optimization

Table 1. Characteristics of different types of models.

Source: Modeling and Simulation of Systems Using MATLAB and Simulink, 2010. p 38

Classifications of Models (Chaturvedi, 2010)

Physical vs. Abstract Model

- Physical models are easy to understand physical replicas, often on a reduced scale. Example: Tabletop scale models

- Abstract models represent systems in terms of logical or quantitative relationships that are manipulated to see specific results. It is composed of symbols that can be in a written language form or a thought process. Example: Flow charts

Mathematical vs. Descriptive Model

- A mathematical model is a subclassification of an abstract model. This type of model encompasses mathematical symbols. Example: The Pythagorean Theorem
- A descriptive model practically describes a system, including its entities and its relationship to the environment. It sometimes serves as a way to explore properties data or information. Example: Model for profiling Google Play

Static vs. Dynamic Model

- Static models represent systems at a specific time or systems wherein time simply plays no role. Example: Floor plans
- Dynamic models represent systems as it evolves over time. This type of model deals with time-varying interactions. Example: Conveyor system model of a factory

Steady State vs. Transient Model

- A steady-state model is consistent with time, wherein the behavior in one specific time is the same as any other period.
- A transient model describes a system with the behavior that changes over time or exhibits growth as it is a one-time phenomenon that cannot recur.

Deterministic vs. Stochastic Models

- Deterministic models do not contain any probabilistic component. In this type of model, the output is determined once the set of input quantities and relationships has been specified. Example: A complex differential equations describing a chemical reaction
- Stochastic models produce an output that is random and must only be treated as an estimate of the true characteristics of the model. Example: An inventory systems model

Continuous vs. Discrete Models

- Whether to use a discrete or continuous model for a particular system depends on the specific objectives of the study.

- In a continuous system model, the system's significant activities or events complete smoothly without any delay. Note that when a continuous system is modeled mathematically, its variables or components representing the attributes are controlled by *continuous functions* (Tutorialspoint, n.d.).
- In a discrete system model, the changes in the system state are discontinuous. Note that the model used in a discrete system simulation has a set of numbers representing the system state called the *state descriptor* (Tutorialspoint, n.d.).

Illustrations of some system models

Example 1:

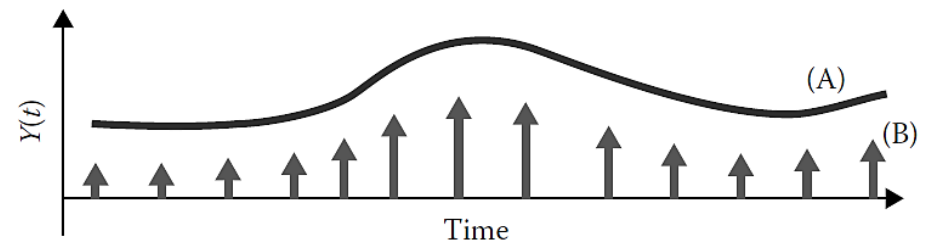


Figure 1. (A) Continuous and (B) Discrete function models

Source: Modeling and Simulation of Systems Using MATLAB and Simulink, 2010. p 38

(A): A model of traffic flow on a freeway, wherein the characteristics and movement of all the vehicles are aggregated. It can be described by differential equations in a continuous model.

(B): A model of traffic flow on a freeway, wherein the characteristics and movement of individual vehicles are significant.

Example 2:

A tank is full of a solution of a material A with a concentration C. A solution of the same material at concentration C_0 is flowing into the tank at flow rate F_0 , and a solution is flowing out of the tank at rate F_1 , as shown in **Figure 2**

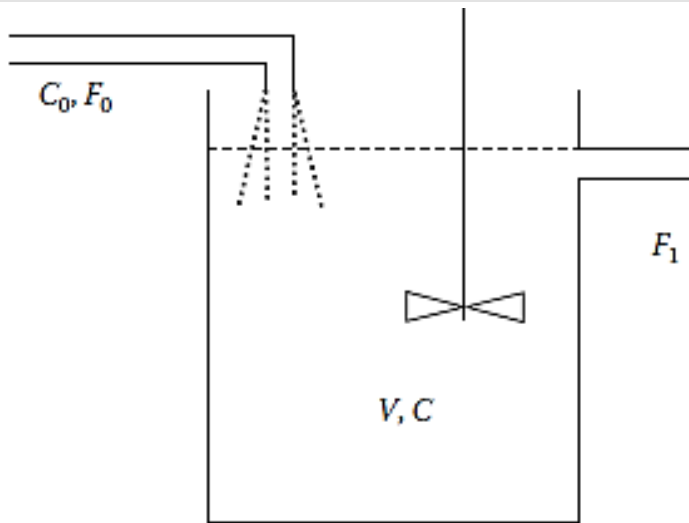


Figure 2. A static model of a hydraulic tank system.

Source: Modeling and Simulation of Systems Using MATLAB and Simulink, 2010. p 44

To determine the dynamic response to a step-change in the inlet concentration C_0 , a set of assumptions, data, and fundamental principles (i.e., the law of conservation of energy and the component terminal equation) must be provided, used, and applied in the mathematical modeling of the given scenario.

The other equivalent models of the given scenario can be generated through manual mathematical computation or modeling and simulation applications, such as the MATLAB® program.

$$C(t) = \bar{C}_0 + (C_{\text{init}} - \bar{C}_0)e^{-\frac{F}{V}t} \text{ when } t \geq 0$$

Figure 3. The mathematical model of the hydraulic tank system.

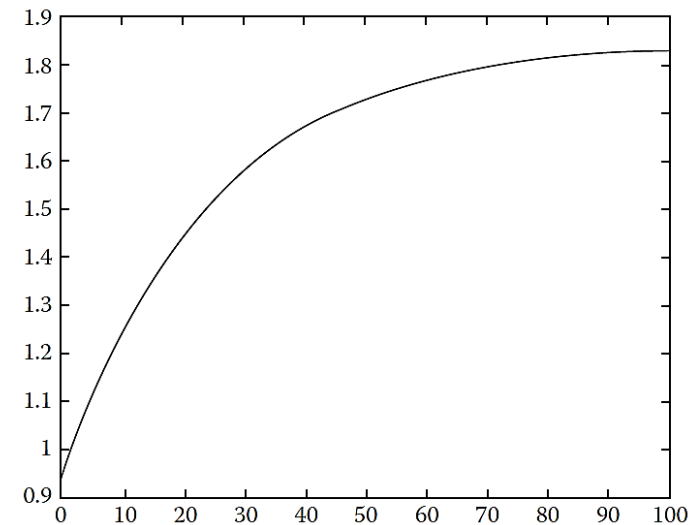


Figure 4. The MATLAB® program generates the dynamic response of the hydraulic tank system.

Source: Modeling and Simulation of Systems Using MATLAB and Simulink, 2010. p 47

References:

- Chaturvedi, D. (2010). *Modeling and simulation of systems using MATLAB and Simulink*. CRC Press – Taylor & Francis Group, LLC
- Dubois, G. (2018). *Modeling and simulation challenges and best practices for industry*. CRC Press – Taylor & Francis Group, LLC
- Sokolowski, J. & Bank, C. (2009). *Principles of modeling and simulation a multidisciplinary approach*. John Wiley & Sons, Inc.
- Tutorialspoint. (n.d.). *Discrete System Simulation*. Retrieve from https://www.tutorialspoint.com/modelling_and_simulation/modelling_and_simulation_discrete_system_simulation.htm on February 4, 2021
- Tutorialspoint. (n.d.). *Modelling & Simulation – Continuous*. Retrieved from https://www.tutorialspoint.com/modelling_and_simulation/modelling_and_simulation_continuous_simulation.htm on February 4, 2021
- Zeigler, B., Muzy, A. & Kofman, E. (2019). *Theory of modeling and simulation discrete event and iterative system computational foundations* (3rd ed.). Academic Press – Elsevier Inc.